Solvency Field Theory

An Information-Theoretic Foundation for Cosmology and Gravity

Part 2: Sections 7–11

Shaun Fosmark

Version 11 — April 2026

7 The Galactic Confirmation

The modified Friedmann equation connects the expansion rate to the matter content through the factor $4\pi^2$. The same factor, applied at galactic scales, predicts a specific acceleration threshold below which gravitational dynamics deviate from Newtonian behavior. This prediction has been tested against 137 galaxies in the SPARC rotation curve sample.

7.1 The acceleration scale

The SFT acceleration scale is

$$\boxed{a_{\mathrm{SFT}} = \frac{cH_0}{2\pi} = 1.04 \times 10^{-10}\ \mathrm{m/s^2}} \tag{17}$$

This is not a free parameter. It is the ratio of the speed of light to the Hubble time, divided by 2π. The galactic acceleration scale carries one leg of the $4\pi^2$ spine directly through the phase/cycle factor 2π, while the full $4\pi^2$ structure appears in the background count normalization and related transfer sectors (Section 8). The acceleration scale is the boundary between two regimes: above a_{SFT}, local structures can absorb their own maintenance costs and gravity behaves as in general relativity; below a_{SFT}, the maintenance cost exceeds the local absorption capacity and the excess is forwarded to the cosmological boundary, producing the anomalous dynamics observed in galaxy rotation curves.

The empirical MOND acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m/s^2 is 13% above the derived a_{SFT}. The residual is flagged as Research Target 3 in Section 15; candidate explanations include QCD binding energy corrections and systematic uncertainties in the empirical determination.

7.2 The interpolating function

At accelerations well above a_{SFT}, the gravitational field matches the Newtonian prediction. At accelerations well below a_{SFT}, the effective gravitational field is enhanced. The transition between these regimes is governed by a completion law derived from the multiplicative forwarding principle:

$$F(x) = 1 - e^{-x}, \qquad x = \frac{g_N}{a_{\mathrm{SFT}}}, \tag{18}$$

where g_N is the Newtonian gravitational acceleration and $F(x)$ is the fraction of maintenance cost absorbed locally. The completion law is not a phenomenological fit; it is derived from the requirement that screening effects compose multiplicatively under nesting (the same composition principle that forces the exponential potential in the action and the exponential metric in the strong-field sector).

The SFT interpolating function for galactic rotation curves takes the form

$$\nu(y) = \sqrt{1 + \frac{\lambda}{y}}, \tag{19}$$

where $y = g_N/a_{\mathrm{SFT}}$ and λ is a geometry-dependent solvency coefficient. The predicted effective gravitational acceleration is $g_{\mathrm{eff}} = g_N \cdot \nu(y)$, so the interpolating function boosts the Newtonian prediction in the low-acceleration regime. The coefficient λ depends on the morphology of the source through the projection geometry described below.

7.3 The SPARC rotation curve test

The interpolating function has been tested against the SPARC (Spitzer Photometry and Accurate Rotation Curves) sample of 137 galaxies with high-quality photometric and kinematic data. The baseline rotation-curve test uses the SFT acceleration scale a_{SFT} fixed by $cH_0/(2\pi)$, with stellar mass-to-light ratios constrained by population models and benchmark solvency coefficients $\lambda_{\mathrm{sph}} = 4/3$ or $\lambda_{\mathrm{disk}} = 3\pi/8$ assigned by morphological class. The result is a 63% reduction in rotation-curve residuals compared to Newtonian-only predictions, with no dark matter halo. The separate morphology diagnostic (Section 7.5) then allows each galaxy's effective λ to vary freely and tests whether the inferred values correlate with morphology in the predicted direction.

7.4 The morphology prediction

The coefficient λ in the interpolating function is not universal. SFT predicts that it depends on the morphology of the source galaxy through a continuous projection family on the two-sphere S^2:

$$M(w, P; c) = c\langle w(\theta)\, P(\theta)\rangle_{S^2}, \tag{20}$$

where the source geometry fixes the participation weight w, the observable fixes the projection kernel P, and the sector equation fixes the prefactor c. This construction writes the galactic response coefficients and the cosmological softening factors (Section 11) as members of one family. Three structural identities link the members across scales.

The unified family generates a continuous morphology curve. Parameterizing the source geometry by a flattening parameter α, with $\alpha = 2$ for an isotropic sphere and $\alpha \to \infty$ for a thin disk, the projection family yields

$$\boxed{\lambda(\alpha) = \sqrt{\pi}\, \frac{\Gamma((\alpha+4)/2)}{\Gamma((\alpha+5)/2)}} \tag{21}$$

interpolating from $\lambda_{\mathrm{sph}} = 4/3$ at the spherical endpoint to $\lambda_{\mathrm{disk}} = 3\pi/8$ at the disk endpoint. (The companion volume uses an equivalent exponent convention $\alpha \in [0, 1]$ in the participation weight $w(\theta; \alpha) = \sin^{\alpha}\theta$; the flattening coordinate used here maps monotonically onto that exponent.)

7.5 Empirical confirmation: the SPARC morphology test

The morphology curve makes a prediction that no other framework in the modified-gravity or dark-matter literature produces: the best-fit solvency coefficient for a galaxy should correlate with the galaxy's morphology, with early-type (bulge-dominated) galaxies landing near 4/3 and late-type (disk-dominated) galaxies landing near $3\pi/8$.

In 131 interior SPARC galaxies, Hubble type T provides the morphology coordinate. The rank correlation with best-fit λ_{obs} is negative and highly significant:

$$\rho(T, \lambda_{\text{obs}}) = -0.3005, \qquad p = 4.87 \times 10^{-4}. \tag{22}$$

The negative sign is the predicted direction: later Hubble types have smaller λ_{obs}, trending from the spherical value toward the disk value. The result remains significant under inclination control:

$$\rho(T, \lambda_{\text{obs}} \mid \text{inclination}) = -0.2732, \qquad p = 0.0016. \tag{23}$$

The early-type endpoint lands at median $\lambda = 1.340$ against the spherical target $4/3 = 1.333$ — a 0.5% offset well within the statistical uncertainty.

This constitutes empirical confirmation of a genuine novel prediction of the framework. To our knowledge, no standard dark-matter model or existing modified-gravity framework predicts this specific Gamma-family morphology curve as a parameter-free projection result. The remaining refinement target is a sharper map from structural photometry to the theory parameter α, replacing the coarse Hubble-type proxy with a continuous morphological variable.

7.6 What this section establishes

The galactic acceleration scale $a_{\text{SFT}} = cH_0/(2\pi)$ is derived, not fitted. The interpolating function is derived from the multiplicative forwarding principle. The morphology-dependent prediction — unique to SFT — is confirmed at $p < 0.001$ with 0.5% endpoint accuracy. The same projection family that produces the galactic coefficients also produces the cosmological corrections of Section 11.

8 The $4\pi^2$ Structural Identity

The factor $4\pi^2$ has appeared twice so far: in the modified Friedmann equation (Section 6) and in the galactic acceleration scale (Section 7, through the factor 2π in $a_{\text{SFT}} = cH_0/(2\pi)$). This section shows that it appears in five distinct physical contexts, and in each context decomposes into the same two independent factors of 2π.

8.1 Five appearances

#	Context	Where $4\pi^2$ enters
1	Modified Friedmann equation	$H^2 = \rho/(4\pi^2)$
2	Galactic acceleration scale	$a_{\text{SFT}} = cH_0/(2\pi)$
3	Holographic count equation	Compton cycles to Planck areas
4	Projection geometry	Same $4\pi^2$ angular family generates cosmological softening and galactic response coefficients
5	CMB transfer coefficient	$\varepsilon_\Phi = 1/(4\pi^2)$, Onsager-derived

These five appearances span four different physical regimes — cosmology, galaxy dynamics, holographic counting, and the early-universe CMB sector — and yet they all produce the same

numerical factor. The structural recurrence is strong evidence for a single underlying mechanism rather than a sequence of coincidences.

8.2 The two-factor decomposition

The full $4\pi^2$ decomposes as the product of two physically independent factors:

$$\boxed{4\pi^2 = (2\pi)_{\text{phase}} \times (2\pi)_{\text{thermal}}} \tag{24}$$

The phase factor. SFT counts one irreversible bit per complete Compton cycle — per h of action, not per $\hbar$. The ratio $h/\hbar = 2\pi$ enters whenever the framework converts between its natural information-counting unit (one complete cycle) and the angular-frequency unit (one radian) used in standard quantum field theory. This factor was already implicit in the Compton Maintenance Postulate; its identification as the first half of the $4\pi^2$ spine is a recognition, not a derivation.

The thermal factor. The second factor is the thermal horizon factor carried by the Unruh temperature in Jacobson's derivation of the Einstein equation from horizon thermodynamics. In Jacobson's construction, the field-equation coupling inherits an explicit factor of 2π through the Unruh temperature $T_U = \hbar a/(2\pi c k_B)$ assigned to local Rindler horizons. The deeper origin is the KMS (Kubo–Martin–Schwinger) periodicity condition: the thermal Green's function of a quantum field in the Rindler vacuum is periodic in imaginary time with period $2\pi/a$, and this periodicity is the structural reason accelerated observers detect a thermal bath. The factor 2π is not a convention; it is a structural fact about the relationship between acceleration and temperature in quantum field theory.

Independence. The two factors are physically independent. The first says how the universe counts maintenance information — in complete cycles rather than radians. The second says how horizons have temperature — through KMS periodicity. Their product appears in five physical contexts because both factors enter wherever quantum phase accounting interfaces with gravitational geometry.

8.3 The Onsager closure of Appearance 5

The candidate transfer coefficient $\varepsilon_\Phi = 1/(4\pi^2)$ that appears in the CMB sector has been derived from a minimal near-equilibrium Onsager interface law. The derivation identifies two unavoidable Jacobians — one from the phase-convention conversion and one from the cycle-angle-to-linear conversion at the relic–photon interface — and shows that the dissipation functional evaluated in the correct Onsager variables produces $\varepsilon_\Phi = 1/(4\pi^2)$ without free parameters. This derivation cleanly separates the bookkeeping normalization (convention-dependent) from the genuine transport block (physics).

8.4 What this section establishes

The factor $4\pi^2$ is not a single unexplained constant. It appears in five distinct physical contexts, in each case decomposing into the same two independent factors of 2π for the same two physical reasons. Both factors have been traced to their independent origins and the decomposition is closed at theorem level. The structural recurrence is strong evidence for a single underlying mechanism: the geometric cost of mapping angular quantum phase evolution onto a linear holographic boundary.

9 The Lagrangian

The reader has now seen the mechanism (Sections 4–5), the modified Friedmann equation it produces (Section 6), the galactic confirmation (Section 7), and the structural identity that recurs across regimes (Section 8). This section presents the field-theoretic action that encodes the framework's established sectors in one variational principle.

9.1 The action

The Lagrangian of Solvency Field Theory is

$$\boxed{S_{\mathrm{SFT}} = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi G} - \eta\,\nabla_\mu(\bar{\chi}+\psi)\,J^\mu_{\mathrm{ops}} + \Lambda_C\,\mathcal{C}_{\min} - V_{\mathrm{quad}} + \mathcal{L}_m\right]} \tag{25}$$

Five terms. Gravity, source coupling, count normalization, potential, and matter. No kinetic term for the bookkeeping scalar. No freely propagating dark-sector field. No free fit parameters once the count normalization $\beta = 1/(4\pi^2)$ is imposed.

9.2 The bookkeeping scalar

The single bookkeeping scalar χ is decomposed as

$$\chi(x) = \bar{\chi}(x) + \psi(x), \tag{26}$$

where $\bar{\chi}$ is the cosmological background clock and ψ is the local lag departure. The field χ is not a new physical field carrying energy through space. It is a bookkeeping variable that tracks the accumulated maintenance debt at each spacetime point. Its gradient defines the local clock direction:

$$u^\mu = \frac{\nabla^\mu(\bar{\chi}+\psi)}{\sqrt{-\nabla_\alpha(\bar{\chi}+\psi)\,\nabla^\alpha(\bar{\chi}+\psi)}}, \qquad u^\mu u_\mu = -1. \tag{27}$$

9.3 Term by term

Term 1: Gravity. The Einstein–Hilbert term $R/(16\pi G)$ is standard general relativity, carried without modification.

Term 2: Source coupling. The term $-\eta\,\nabla_\mu\chi\,J^\mu_{\mathrm{ops}}$ couples the gradient of the bookkeeping scalar to the covariant operation-density current $J^\mu_{\mathrm{ops}} = \rho_{\mathrm{ops}}\,u^\mu$, where $\rho_{\mathrm{ops}} = (\rho+3p)/h$ is the operation density derived in Section 5. This is the term that connects matter's information-processing load to the geometric response. The coupling constant η is fixed by the requirement that the FLRW radiation reduction produces the correct asymptotic normalization $\beta = 1/(4\pi^2)$.

The source coupling is lapse-independent in FLRW: $\partial\mathcal{L}_{\mathrm{src}}/\partial N = 0$. It does not enter the Hamiltonian constraint as an additional fluid. It is a driver, not a tank.

Term 3: Count normalization. The term $\Lambda_C\,\mathcal{C}_{\min}$ enforces the holographic count law at the local covariant level:

$$\mathcal{C}_{\min} = \mathcal{H}_\chi^2 - \alpha\big(E_\chi + V_{\mathrm{quad}}\big), \tag{28}$$

with

$$\mathcal{H}_\chi = \tfrac{1}{3}\nabla_\mu u^\mu, \qquad E_\chi = T_{\mu\nu}u^\mu u^\nu, \qquad \alpha = \tfrac{8\pi G}{3}\beta, \qquad \beta = \frac{1}{4\pi^2}. \tag{29}$$

The constraint $\mathcal{C}_{\min} = 0$ is the local statement that expansion rate (boundary growth) balances information-maintenance demand. In homogeneous FLRW, $\mathcal{H}_\chi = H$ and $E_\chi = \rho$, and the constraint recovers $H^2 = \rho/(4\pi^2)$. The count constraint is the only term carrying the $4\pi^2$ normalization in the dust sector, where the source coupling vanishes.

Term 4: Potential. The quadratic background-relative potential

$$V_{\text{quad}} = V_0(e^{-\psi} - 1)^2 \tag{30}$$

measures the local departure from the cosmological background. It satisfies $V_{\text{quad}}|_{\psi=0} = 0$, $dV_{\text{quad}}/d\psi|_{\psi=0} = 0$, and $d^2V_{\text{quad}}/d\psi^2|_{\psi=0} = 2V_0 > 0$: zero value, zero force, positive restoring curvature on the background branch.

Term 5: Matter. The standard matter Lagrangian $\mathcal{L}_m$ of general relativity, unmodified. Although $\mathcal{L}_m$ itself is standard, the SFT bookkeeping sector couples to the stress-energy tensor derived from it through the projected operation current J^μ_{ops}; the variation of this dependence is handled in the companion volume.

9.4 Why this action and not another

The action did not appear fully formed. It survived a systematic elimination process.

An earlier candidate carried an absolute exponential potential $V_0 e^{-\chi}$. This potential obstructed exact static vacuum: on a static Killing-aligned clock branch, $V_0 e^{-\chi}$ survives as a time-dependent source even when all other SFT-specific terms vanish. The repaired quadratic background-relative potential is the first candidate satisfying both zero value and zero force on the background branch, which is required for exact vacuum decoupling.

An earlier candidate carried two independent scalar fields with a coupled potential designed to drive the bridge ratio toward $1/(2\pi)$ dynamically. The scalar sector dominated the Hamiltonian — the bookkeeping fields became a runaway energy reservoir. The two-field branch was rejected for this specific physical reason.

An earlier candidate retained a kinetic term $(\varepsilon/2)\nabla_\mu\chi\nabla^\mu\chi$. The honest χ-variation of the full action — carrying the implicit χ-dependence through u^μ, ρ_{ops}, and J^μ_{ops} — showed that the hidden frame-rotation contribution is controlled entirely by the clock-frame spatial momentum flux q^μ, and vanishes exactly in all comoving perfect-fluid sectors. No established result requires $\varepsilon \neq 0$. The kinetic term was removed.

Each elimination was forced by a specific failure, and the surviving action incorporated the structural lesson. The branch genealogy — from Branch A through Branch C* to the single-field action — is documented in the companion volume.

9.5 What the action produces

Exact Schwarzschild exterior. On the background branch $\psi = 0$ in static vacuum ($T_{\mu\nu} = 0$): $V_{\text{quad}} = 0$, $dV_{\text{quad}}/d\psi = 0$, $J^\mu_{\text{ops}} = 0$, $E_\chi = 0$, $\mathcal{H}_\chi = 0$ (static spacetime has zero expansion), and $\mathcal{C}_{\min} = 0$ (automatically satisfied). Every SFT-specific term vanishes simultaneously. The action reduces to pure Einstein–Hilbert:

$$S_{\text{vac}} = \int d^4x\,\sqrt{-g}\,\frac{R}{16\pi G}. \tag{31}$$

The vacuum field equation is $G_{\mu\nu} = 0$, and the unique static, spherically symmetric, asymptotically flat solution is the Schwarzschild metric. This recovery is exact, not approximate.

Modified Friedmann equation. In FLRW dust, the source coupling vanishes and the count constraint alone delivers $H^2 = \rho/(4\pi^2)$, exactly. In FLRW radiation, the source coupling is active and the background equation is asymptotically exact at $\beta = 1/(4\pi^2)$.

Vacuum decoupling. With $\rho = 0$ and $\psi = 0$, the count constraint gives $H = 0$. No residual SFT vacuum load appears on this branch.

Modified TOV interior (branch-level/effective-fluid closure). Inside a static spherical matter distribution on the $\psi = 0$ branch:

$$m'(r) = 4\pi r^2 (1 + \Lambda_C \alpha)\, \rho(r), \tag{32}$$

$$\Phi'(r) = \frac{m(r) + 4\pi r^3 \Big[(1 + \Lambda_C \alpha)\, p - \dfrac{\eta\kappa}{h}\, e^{-\Phi}(\rho + 3p)\Big]}{r[r - 2m(r)]}, \tag{33}$$

$$p'(r) = -(\rho + p)\, \Phi'(r). \tag{34}$$

The count sector rescales enclosed mass. The source sector adds a local correction to radial pressure. The center is regular. The Newtonian limit is clean.

9.6 What this section establishes

The Lagrangian of Solvency Field Theory is a five-term action encoding the framework's established sectors in one variational principle. It recovers exact Schwarzschild in static vacuum, exact dust normalization, asymptotically exact radiation normalization, vacuum decoupling, and a branch-level modified TOV interior with an exact mass equation. The action arrived through systematic elimination of failed branches, not by construction. Its remaining open items — background-independent packaging of $\bar{\chi}$, full variational closure of the radial sector, and interior phenomenology — are carry-the-derivation-further tasks, not conceptual gaps.

10 Forced Thermodynamic Transitions of the Maintenance Process

Sections 4 through 9 established what the maintenance process is and what it requires at any given moment. This section asks: what happens when the maintenance process can no longer sustain its current regime?

10.1 The principle

A forced thermodynamic transition occurs when the dominant maintenance channel can no longer service the information-processing demand of the universe. The transition is not optional — the system must reorganize onto a different channel, and the reorganization leaves observable scars on the cosmic history.

10.2 The unified solvency ratio

Both known transitions — the early-universe shatter and the late-universe engine crash — are described by a single critical variable:

$$\boxed{\Lambda_X(t) = \frac{D(t)}{C_X(t)} = \frac{H(t)\,I(t)}{C_X(t)}} \tag{35}$$

where $D(t) = H(t)\,I(t)$ is the total maintenance demand (expansion rate times total information content) and $C_X(t)$ is the capacity of channel X to service that demand. The orientation is sign-consistent:

$$\Lambda < 1 \text{ (solvent)}, \qquad \Lambda = 1 \text{ (critical)}, \qquad \Lambda > 1 \text{ (insolvent)}. \tag{36}$$

A forced transition occurs when Λ_X crosses unity. The shatter is the crossing at the shared→independent boundary. The engine crash is the crossing at the active→passive boundary.

10.3 The four-regime maintenance history

The corrected solvency ratio reveals that the maintenance history of the universe is a four-regime sequence:

$$\text{shared} \;\rightarrow\; \text{independent} \;\rightarrow\; \text{active} \;\rightarrow\; \text{passive}. \tag{37}$$

The shared regime. The earliest phase. The radiation-dominated plasma maintains all particles through a single shared bookkeeping system. As the universe cools and massive particles demand independent Compton maintenance, the shared channel weakens. The solvency ratio crosses unity at the *shatter*.

The independent regime. Each massive particle now maintains itself through its own Compton phase evolution. Astrophysical entropy engines have not yet organized into the dominant channel. An independent or seed-structure bridge channel carries the universe through this intermediate epoch — without it, the maintenance envelope leaves a post-shatter insolvent gap.

The active regime. Astrophysical entropy engines — star formation, black hole accretion — become the dominant servicing channel. This regime peaks near $z \approx 2$ and declines as fuel is exhausted.

The passive regime. The active channel can no longer service the demand, and the system transitions to a structure-dominated regime in which the passive occupancy cost of organized gravitational architecture dominates the maintenance balance. This is the regime the universe currently occupies.

10.4 Robustness

The four-regime picture is not a knife-edge. It occupies a real basin in parameter space. A pinned effective sweep yields 489 accepted models out of 972 tested (acceptance fraction 0.503). A softer local-release sweep preserves 1836 accepted models out of 8748 tested (acceptance fraction 0.210). The marginal sensitivity ranking identifies the shared-channel weakening as the most fragile direction and the passive channel as the most tolerant.

The honest status is: the corrected solvency ratio and the four-regime sequence are established at effective closure level with a real robustness basin. The remaining frontier is the first-principles derivation of the four channel histories from the axioms, rather than from effective parameterization.

10.5 What this section establishes

The shatter and the engine crash are not two separate phenomena that happen to look similar. They are two channel-critical crossings of one solvency ratio under different dominant servicing channels. The four-regime maintenance history is the minimal sequence consistent with the requirement that the dominant maintenance envelope remain solvent throughout the universe's history. The robustness basin is real.

11 The Cosmological Sector

The unified solvency ratio describes when the maintenance process transitions. This section describes what the late-universe transition produces: the dark energy equation of state and the observer-side environment correction.

11.1 The four-state cosmic web ladder

The large-scale structure of the universe is not uniform. Matter is organized into voids, walls, filaments, and halos — a cosmic web whose geometry determines the local maintenance environment. In SFT, these four environments are characterized by their dimensionality and their ability to absorb maintenance costs locally.

The four-state ladder assigns environmental weights through an axis-factorization principle derived from three axioms (C1: each axis contributes independently, C2: the contribution is binary — collapsed or uncollapsed, C3: a fully collapsed environment has maximum local absorption). The resulting weights for the four environments determine their contributions to the effective dark energy equation of state.

11.2 The unified projection family

The static projection corrections for the cosmic web environments are not independent phenomenological parameters. They are members of the same unified projection family on S^2 that produces the galactic morphology curve of Section 7:

$$M(w, P; c) = c\langle w(\theta)\, P(\theta)\rangle_{S^2}. \tag{38}$$

The cosmological softening factors ($s_{\text{wall}} = 1/3$, $s_{\text{filament}} = 2/3$) and the galactic response coefficients ($\lambda_{\text{sph}} = 4/3$, $\lambda_{\text{disk}} = 3\pi/8$) are four evaluations of one construction. Three structural identities — normalization, kernel, and prefactor — link the members across scales. This cross-scale unification is closed at theorem level.

11.3 The observer-side shell correction

The observer's local cosmic-web environment introduces a correction to the effective unscreened burden at each redshift. The multiplicative forwarding law — screening effects compose multiplicatively through nested structures — yields the observer-side shell correction:

$$\Delta U_{\text{rec}}(z) = -\frac{1}{\chi(z)} \sum_i f_{\text{cov},i}\, [U_i - U_{\text{global}}(z)]\, L_i(z), \tag{39}$$

where $f_{\text{cov},i}$ is the covering fraction of structure i on the observer's sky, U_i is its screening level, $U_{\text{global}}(z)$ is the global background at redshift z, and $L_i(z)$ is the path length. This formula is

the first-order sky/path expansion of the same multiplicative composition principle that forces the exponential potential in the action and the exponential metric in the strong-field sector.

11.4 The dark energy equation of state

The historical reconstruction pipeline combines the four-state ladder, the static projection corrections, and the channel-history weights to produce a dark energy equation of state $w(z)$ from first principles. The derived $w(z)$ naturally crosses $w = -1$ at $z \approx 0.3$–0.5, matching the DESI DR2 measurement without free parameters. The crossing is not introduced by hand — it is forced by the thermodynamic transition from the active to the passive maintenance regime (the engine crash). The current status is pipeline-level reconstruction agreement with DESI DR2 features; the CPL coefficient reconciliation under the corrected chain remains the highest-priority audit item (Research Target 1 in Section 15).

11.5 What this section establishes

The late-universe cosmological sector produces a dark energy equation of state matching current observations through a combination of cosmic-web environmental weights, projection corrections from the unified screen family, and the forced thermodynamic transition of the maintenance process. The projection corrections are derived from the same construction that produces the galactic morphology curve, providing a cross-scale structural confirmation. The observer-side environment correction has a specific mathematical form derived from the multiplicative forwarding law.

End of Part 2 — Sections 7 through 11. Part 3 continues with the early-universe sector (Section 12), the strong-field closure (Section 13), predictions (Section 14), open theorems (Section 15), and conclusion (Section 16).

Portions of the mathematical derivation and manuscript preparation were developed in collaboration with Claude/Opus (Anthropic), ChatGPT/5.4 (OpenAI), Gemini/Ender (Google), and Grok (xAI).